Answer Key & Teacher Notes

Session 1 — Percent Means Per Hundred · Session 2 — Percents Greater Than 100%

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages: 8 problems, Find the Error, 2 test items. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Teaching Support Guide	Teacher page: lesson route, what to look for on each exit item, the next prompt, and a fresh check for after reteaching.
Explain Your Thinking	Optional language support for the Guided Notes "Model, Reason, Explain" page: draw it, talk it, write it.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank (English and Spanish), step frames on every problem and larger type (12 pt).

SESSION 1 · TEACHER NOTES

Percent Means Per Hundred

TALK PROMPTS

- Why can we compare the camp choices as percents even though only 25 campers were counted?
- Why must the coupon percent and the no-coupon percent together make 100%?

PARTNER TASK

Coupons, Grid Shading, Sneaker Survey. One partner scales each ratio to a per-hundred ratio; the other shades a decimal grid to check the percent. Switch roles for each problem.

WATCH FOR

- Writing the part as the percent when the whole is not 100 (11 out of 25 = 11%).
- Adding to reach 100 (11 + 80 out of 20 + 80) instead of multiplying both terms.
- Forgetting that the part and the rest must add to 100%.

SESSION 2 · TEACHER NOTES

Percents Greater Than 100%

TALK PROMPTS

- How can a percent be greater than 100 when a decimal grid only holds 100 squares?
- Friday's jump-rope percent came out greater than 100. What does that tell you about the two counts?

PARTNER TASK

Practice Minutes, Jump Rope, Video Views. Partners take turns naming the whole, finding 10% of it, and counting up to the other amount. Then switch which amount is the whole and compare the two percents.

WATCH FOR

- Using the wrong amount as the whole (Monday as a percent of Tuesday instead of Tuesday of Monday).
- Writing the amount itself as the percent (16 kg is 16%).
- Thinking a percent can never be more than 100% (writing 30 cm compared to 12 cm as 30%).

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.AT.4	1	2	1	1	1	2	2	3	3	1	1
S1 · Form B	6.AT.4	1	2	1	1	1	2	2	3	3	1	1
S2 · Form A	6.AT.4	1	2	2	2	1	2	2	3	3	1	1
S2 · Form B	6.AT.4	1	2	2	2	1	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

percent: Out of **100**, or per hundred.

decimal grid: A grid of 100 squares. Each square is **1%**.

equivalent ratio: Shows the **same** comparison with different numbers.

- Percent means per **hundred**. The whole is always **100%**.
- Multiply the part and the whole by the **same** number to make the whole 100.

Example: What times 4 makes 100? **25**. Multiply both: 3 × 25 = **75** and 4 × 25 = 100. 75 out of 100 = 75%. Answer: **75%** of the students bring lunch.

YOU TRY

1. a. 30% b. 70%

2. a. 90% b. 35%

EXIT TICKET

1. a. 80% b. 38%

2. a. 6 × 4 = 24, so 24% b. 100% − 24% = 76%

3. **C** (60%)

PRACTICE FORM A

1. a. 75% b. 15% c. 42% d. 40% e. 32% f. 85%

2. Each count × 4: 56%, 28%, 16%

3. a. 62% b. 0.62

4. a. 11 × 5 = 55, so 55% b. 100% − 55% = 45%

5. 33 × 2 = 66 out of 100, so 66%

6. No. 12 out of 25 = 48 out of 100, so 48%.

7. 25 − 16 = 9 moon patches; 9 × 4 = 36 out of 100, so 36%

8. a. 26 out of 40 = 13 out of 20 = 65 out of 100, so 65% b. 35% c. The two groups make the whole, so 65% + 35% must equal 100%.

9. Her whole is 25, not 100, so 11 is not a percent yet. Scale to 100: 11 × 4 = 44, so 44% chose soccer.

10. **D** (per hundred)

11. **A** (56%)

Self-check word **PERCENT**

PRACTICE FORM B

1. a. 25% b. 65% c. 58% d. 60% e. 24% f. 95%

2. Each count × 5: 45%, 30%, 25%

3. a. 37% b. 0.37

4. a. 17 × 5 = 85, so 85% b. 100% − 85% = 15%

5. 27 × 2 = 54 out of 100, so 54%

6. No. 18 out of 25 = 72 out of 100, so 72%.

7. 20 − 13 = 7 moon patches; 7 × 5 = 35 out of 100, so 35%

8. a. 22 out of 40 = 11 out of 20 = 55 out of 100, so 55% b. 100% − 55% = 45% c. The two groups make the whole, so 55% + 45% must equal 100%.

9. Her whole is 25, not 100, so 9 is not a percent yet. Scale to 100: 9 × 4 = 36, so 36% sprouted.

10. **B** (45%)

11. **C** (100)

Self-check word **HUNDRED**

GUIDED NOTES

the whole: The amount that counts as **100%**.

times as much: Tells how many **wholes** an amount makes.

greater than 100%: More than the whole.

- 100% is one **whole**. More than 100% means more than one whole.
- Always name the **whole** first. Then compare the other amount to it.

Example: The whole: 5 pounds = **100%**. 1 pound = 100% ÷ 5 = **20%**. 8 pounds = 8 × 20% = 160%. Answer: 8 pounds is **160%** of last month's weight.

YOU TRY

1. a. 200% b. 160%

2. 45 to 30 = 150 to 100, so 150%

EXIT TICKET

1. a. 450% b. 125%

2. 36 ÷ 24 = 1.5, so 150%

3. **A** (30)

PRACTICE FORM A

1. a. 400% b. 150% c. 250% d. 120%

2. Missing: 40%, 60%, 80%, 120%, 140%, 160%. This year is 160% of last year.

3. a. 50 to 40 = 125 to 100, so 125% b. 40 to 50 = 80 to 100, so 80%

4. 90 to 50 = 180 to 100, so 180%

5. 1,100 to 500 = 220 to 100, so 220%

6. No. The new batch is 2 whole batches, so it is 200% of the first batch.

7. Missing: 25%, 50%, 75%, 125%, 150%, 175%. 18 ÷ 12 = 1.5, so 150%

8. a. 30 ÷ 12 = 2.5, so 250% b. 12 to 30 = 2 to 5 = 40 to 100, so 40% c. In a, May (12 cm) is the whole and July is taller, so July is more than 100%. In b, July (30 cm) is the whole and May is shorter, so May is less than 100%.

9. 16 is the weight, not the percent. The small pumpkin (10 kg) is the whole: 16 to 10 = 160 to 100, so the large pumpkin is 160% of the small one.

10. **B** (300%)

11. **D** (120%)

Self-check word **GREATER**

PRACTICE FORM B

1. a. 500% b. 140% c. 350% d. 110%

2. Missing: 40%, 60%, 80%, 120%, 140%, 160%, 180%. This year is 180% of last year.

3. a. 40 to 32 = 125 to 100, so 125% b. 32 to 40 = 80 to 100, so 80%

4. 85 to 50 = 170 to 100, so 170%

5. 1,000 to 400 = 250 to 100, so 250%

6. No. The new batch is 3 whole batches, so it is 300% of the first batch.

7. Missing: 25%, 50%, 75%, 125%, 150%, 175%. 20 ÷ 16 = 1.25, so 125%

8. a. 36 ÷ 9 = 4, so 400% b. 9 to 36 = 1 to 4 = 25 to 100, so 25% c. In a, May (9 cm) is the whole and July is taller, so July is more than 100%. In b, July (36 cm) is the whole and May is shorter, so May is less than 100%.

9. 18 is the weight, not the percent. The small pumpkin (12 kg) is the whole: 18 ÷ 12 = 1.5, so the large pumpkin is 150% of the small one.

10. **C** (more than)

11. **A** (1.5 times it)

Self-check word **AMOUNTS**

SHEET 1 · PERCENT MEANS PER HUNDRED

1. a. 46% b. 54%
2. a. 30% b. 28% c. 50% d. 34% e. 40% f. 15% g. 52% h. 90%
3. Missing: 27, 36. 9 out of 25 = 36 out of 100 = 36%.
4. a. 22 × 2 = 44 out of 100 = 44% b. 100% − 44% = 56%
5. Equivalent ratios are made by multiplying (or dividing) both terms by the same number, not adding. 20 × 5 = 100, so 7 × 5 = 35: 7 out of 20 = 35%.

SHEET 2 · PERCENTS GREATER THAN 100%

1. a. 300% b. 175%
2. Missing percents: 50%, 75%, 125%, 150%, 175%. Tuesday = 175% of Monday.
3. 260 to 200 = 130 to 100, so 130% (more than the goal).
4. 33 to 20 = 165 to 100 (multiply both by 5), so 165%.
5. 60% of 30 cm is only 18 cm, which is less than 30 cm. 60 is 2 times 30, so 60 cm is 200% of 30 cm.
6. **D** (150%)
7. **B** (2.5 times)

SESSION 1 · PERCENT MEANS PER HUNDRED

a. 3, 6, 9; shade 3 of the 4 parts. b. 75% c. All 12 challenges are the whole, 100%. Each part is 3 challenges, or 25%. 9 challenges fill 3 parts, so 3 × 25% = 75%.

Challenge One part of the bar is already 25%, so 9% would be less than one part. 9 challenges fill 3 of the 4 parts, so the team finished 75%, not 9%.

SESSION 2 · PERCENTS GREATER THAN 100%

a. 25%, 50%, 75%, 125%, 150%, 175%; shade 7 of the 8 parts. b. 175% c. The gray kitten (8 ounces) is the whole, 100%. Each part is 2 ounces, or 25%. 14 ounces is 7 parts, so 7 × 25% = 175%. The brother is more than one whole, so the percent is more than 100%.

Challenge 14% is less than one part of the bar (25%, or 2 ounces). The brother is heavier than the gray kitten, so his weight must be more than 100%. He fills 7 parts: 175%.

Score each task 0–3: 1 for a model that fits the numbers, 1 for correct math with units, 1 for an explanation that matches the model.